

# CHUAN MENG

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## RESEARCH INTERESTS

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My research lies at the intersection of Agentic AI, Information Retrieval (IR), and Natural Language Processing (NLP), with a focus on **cost-effective long-horizon agents for information seeking**, such as deep research agents.

## WORK EXPERIENCE

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### The University of Edinburgh, United Kingdom

*September 2025 - present*

Postdoctoral Researcher (Research Associate)

Funded by Turing AI Fellowships

Supervisor: Dr. [Jeff Dalton](#)

### Amazon, United Kingdom

*August 2024 - January 2025*

Applied Scientist Intern, working on proactive conversational agents

Manager: Dr. [Gabriella Kazai](#). Mentor: Dr. [Francesco Tonolini](#).

## EDUCATION

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### University of Amsterdam, The Netherlands

*October 2021 - June 2025*

Ph.D. in Artificial Intelligence

Supervisors: Prof. dr. [Maarten de Rijke](#) (ACM Fellow) and dr. [Mohammad Aliannejadi](#)

PhD thesis: Optimizing Agentic Workflows for Information Access [\[pdf\]](#)

### Shandong University, China

*September 2018 - June 2021*

Master's degree in Computer Science and Technology

Supervisors: Prof. dr. [Zhumin Chen](#), dr. [Pengjie Ren](#), and dr. [Zhaochun Ren](#)

Master's thesis: Research on Knowledge-Grounded Non-Task-Oriented Conversational System [\[pdf\]](#)

### Shandong Normal University, China

*September 2014 - June 2018*

Bachelor's degree in Electronic Commerce

## PUBLICATIONS

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As of September 2026, I have 800+ citations (Google Scholar) with an h-index of 17. I have authored papers published in proceedings/journals, such as ACL, EMNLP, NAACL, SIGIR, CIKM, AAAI, ECIR, and TOIS.

### 1. Revisiting Text Ranking in Deep Research

**Chuan Meng**, Litu Ou, Sean MacAvaney, and Jeff Dalton

**SIGIR 2026**: The 49th International ACM SIGIR Conference on Research and Development in Information Retrieval

[\[pdf\]](#), [code](#)

### 2. Agent and Tool Search: Foundations, Techniques, and Open Challenges

Bin Wu, Hossein A. Rahmani, To Eun Kim, Arastun Mammadli, Shuofei Qiao, Xiao Fu, Varsha Ramineni, Xiaoyu Zhang, **Chuan Meng**, George Drayson, Anu Chowdhury, Jerome Ramos, Abdine Maiga, and Emine Yilmaz

Under review

[\[pdf\]](#)

### 3. Re-Rankers as Relevance Judges

**Chuan Meng**, Jiqun Liu, Mohammad Aliannejadi, Fengran Mo, Jeff Dalton, and Maarten de Rijke

Under review

[\[pdf\]](#), [code](#)

4. **Toward Exploring Mixed-Initiative Conversation Generation Based on Community Question Answering**  
Lili Lu, Pranav Kasela, Federico Ravenda, **Chuan Meng**, Gabriella Pasi, and Fabio Crestani  
**TOIS**: ACM Transactions on Information Systems, 44(5), 1–29, 2026  
[\[pdf\]](#)
5. **Generative Retrieval with Few-shot Indexing**  
Arian Askari\*, **Chuan Meng**\* (co-first author), Mohammad Aliannejadi, Zhaochun Ren, Evangelos Kanoulas, and Suzan Verberne  
**ECIR 2026**: The 48th European Conference on Information Retrieval  
[\[pdf\]](#)
6. **Towards Adaptive Personalized Conversational Information Retrieval**  
Fengran Mo, Yuchen Hui, Yuxing Tian, Zhaoxuan Tan, **Chuan Meng**, Zhan Su, Kaiyu Huang, Jian-Yun Nie  
**CIKM 2025**: The 34th ACM International conference on Information and Knowledge Management  
[\[pdf\]](#)
7. **UniConv: Unifying Retrieval and Response Generation for Large Language Models in Conversations**  
Fengran Mo, Yifan Gao, **Chuan Meng**, Xin Liu, Zhuofeng Wu, Kelong Mao, Zhengyang Wang, Pei Chen, Zheng Li, Xian Li, Bing Yin, Meng Jiang  
**ACL 2025**: The 63rd Annual Meeting of the Association for Computational Linguistics  
[\[pdf\]](#)
8. **Bridging the Gap: From Ad-hoc to Proactive Search in Conversations**  
**Chuan Meng**, Francesco Tonolini, Fengran Mo, Nikolaos Aletras, Emine Yilmaz, and Gabriella Kazai  
**SIGIR 2025**: The 48th International ACM SIGIR Conference on Research and Development in Information Retrieval  
[\[pdf\]](#), [code](#)
9. **Query Performance Prediction using Relevance Judgments Generated by Large Language Models**  
**Chuan Meng**, Negar Arabzadeh, Arian Askari, Mohammad Aliannejadi, and Maarten de Rijke  
**TOIS**: ACM Transactions on Information Systems, 43(4), 1–35, 2025  
[\[pdf\]](#), [code](#)
10. **SOLID: Self-seeding and Multi-intent Self-instructing LLMs for Generating Intent-aware Information-Seeking Dialogs**  
Arian Askari, Roxana Petcu, **Chuan Meng**, Mohammad Aliannejadi, Amin Abolghasemi, Evangelos Kanoulas, and Suzan Verberne.  
**NAACL 2025**: The 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics  
[\[pdf\]](#), [code](#)
11. **Zero-Shot and Efficient Clarification Need Prediction in Conversational Search**  
Lili Lu, **Chuan Meng**, Federico Ravenda, Mohammad Aliannejadi and Fabio Crestani  
**ECIR 2025**: The 47th European Conference on Information Retrieval  
[\[pdf\]](#), [code](#)
12. **Improving the Re-Usability of Conversational Search Test Collections**  
Zahra Abbasiantaeb, **Chuan Meng**, Leif Azzopardi and Mohammad Aliannejadi  
**ECIR 2025**: The 47th European Conference on Information Retrieval  
[\[pdf\]](#), [code](#)
13. **Can We Use Large Language Models to Fill Relevance Judgment Holes?**  
Zahra Abbasiantaeb, **Chuan Meng**, Leif Azzopardi, Mohammad Aliannejadi  
**LLM4Eval**: The First Workshop on Large Language Models (LLMs) for Evaluation in Information Retrieval  
[\[pdf\]](#), [code](#)
14. **Ranked List Truncation for Large Language Model-based Re-Ranking**

- Chuan Meng**, Negar Arabzadeh, Arian Askari, Mohammad Aliannejadi, and Maarten de Rijke.  
**SIGIR 2024**: The 47th International ACM SIGIR Conference on Research and Development in Information Retrieval  
[\[pdf\]](#), [code](#)
15. **Query Performance Prediction for Conversational Search and Beyond**  
**Chuan Meng**  
**SIGIR 2024**: The 47th International ACM SIGIR Conference on Research and Development in Information Retrieval  
[\[pdf\]](#)
  16. **LLM-based Retrieval and Generation Pipelines for TREC Interactive Knowledge Assistance Track (iKAT) 2023.**  
Zahra Abbasiantaeb, **Chuan Meng**, David Rau, Antonis Krasakis, Hossein A. Rahmani, and Mohammad Aliannejadi.  
**TREC 2023**: The Thirty-Second Text REtrieval Conference (**Our submitted runs ranked 1st**)  
[\[pdf\]](#)
  17. **Expand, Highlight, Generate: RL-driven Document Generation for Passage Reranking**  
Arian Askari, Mohammad Aliannejadi, **Chuan Meng**, Evangelos Kanoulas, and Suzan Verberne  
**EMNLP 2023**: The 2023 Conference on Empirical Methods in Natural Language Processing  
[\[pdf\]](#), [code](#)
  18. **System Initiative Prediction for Multi-turn Conversational Information Seeking**  
**Chuan Meng**, Mohammad Aliannejadi, and Maarten de Rijke  
**CIKM 2023**: The 32nd ACM International Conference on Information and Knowledge Management  
[\[pdf\]](#), [code](#)
  19. **Query Performance Prediction: From Ad-hoc to Conversational Search**  
**Chuan Meng**, Negar Arabzadeh, Mohammad Aliannejadi, and Maarten de Rijke  
**SIGIR 2023**: The 46th International ACM SIGIR Conference on Research and Development in Information Retrieval  
[\[pdf\]](#), [code](#)
  20. **Performance Prediction for Conversational Search Using Perplexities of Query Rewrites**  
**Chuan Meng**, Mohammad Aliannejadi, and Maarten de Rijke  
**QPP++ 2023**: Query Performance Prediction and Its Evaluation in New Tasks Workshop co-located with The 45th European Conference on Information Retrieval  
[\[pdf\]](#), [code](#)
  21. **Initiative-Aware Self-Supervised Learning for Knowledge-Grounded Conversations**  
**Chuan Meng**, Pengjie Ren, Zhumin Chen, Zhaochun Ren, Tengxiao Xi, and Maarten de Rijke  
**SIGIR 2021**: The 44th International ACM SIGIR Conference on Research and Development in Information Retrieval  
[\[pdf\]](#)
  22. **Conversations Powered by Cross-Lingual Knowledge**  
Weiwei Sun<sup>\*</sup>, **Chuan Meng**<sup>\*</sup> (co-first author), Qi Meng, Zhaochun Ren, Pengjie Ren, Zhumin Chen, and Maarten de Rijke  
**SIGIR 2021**: The 44th International ACM SIGIR Conference on Research and Development in Information Retrieval  
[\[pdf\]](#), [code](#)
  23. **DukeNet: A Dual Knowledge Interaction Network for Knowledge-Grounded Conversation**  
**Chuan Meng**, Pengjie Ren, Zhumin Chen, Weiwei Sun, Zhaochun Ren, Zhaopeng Tu, and Maarten de Rijke  
**SIGIR 2020**: The 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval  
[\[pdf\]](#), [code](#)
  24. **RefNet: A Reference-aware Network for Background Based Conversation**  
**Chuan Meng**, Pengjie Ren, Zhumin Chen, Christof Monz, Jun Ma, and Maarten de Rijke

## ACADEMIC SERVICE

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So far, I have organized a workshop, served on the program committees of 13 major venues (including as a senior program committee member), reviewed for 3 journals, and served as a moderator for arXiv.

- Workshop organization:
  - **QPP++2025: Query Performance Prediction and its Applications in the Era of Large Language Models**  
**Chuan Meng**, Guglielmo Faggioli, Mohammad Aliannejadi, Nicola Ferro, and Josiane Mothe  
**ECIR 2025**: The 47th European Conference on Information Retrieval  
6–10 April 2025, Lucca, Italy  
[\[pdf, website\]](#)
- Senior Program Committee member: CIKM 2026
- Program Committee member: ACL 2023, EMNLP 2023–2021, SIGIR 2026–2024, COLING 2025, 2022 & 2020, CIKM 2025–2022, The Web Conf 2026–2024, WSDM 2026–2022, SIGIR-AP 2026, ECIR 2026–2024, ICTIR 2025 & 2023, SIGKDD 2022, ECML/PKDD 2022–2021, AAAI 2021
- ACL Rolling Review (ARR) reviewer 2025
- Journal reviewer: Transactions on Information Systems (TOIS), Pattern Recognition, Information Processing and Management (IP&M)
- arXiv, Moderator (cs.IR), February 2026–present

## TEACHING & SUPERVISION

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- Tutorial organization: so far, I have organized 7 tutorials, as the first speaker in 3 of them and the second speaker in the other 4.
  - **Agentic Search: Foundations, Optimization, and Long-Horizons**  
**Chuan Meng**, Litu Ou, Sean MacAvaney, and Jeff Dalton  
**CIKM 2026**: The 35th ACM International Conference on Information and Knowledge Management (upcoming)  
7 November 2026, Rome, Italy
  - **Conversational Search: From Fundamentals to Frontiers in the Age of Agents**  
**Chuan Meng**, Fengran Mo, Mohammad Aliannejadi, Jeff Dalton, and Jian-Yun Nie  
**WWW 2026**: The ACM Web Conference 2026  
30 June 2026, Dubai, United Arab Emirates  
[\[paper, slides, website\]](#)
  - **Conversational Search: Foundations, Large Language Models, and Agents**  
**Chuan Meng**, Fengran Mo, Mohammad Aliannejadi, Jeff Dalton, and Jian-Yun Nie  
**ECIR 2026**: The 48th European Conference on Information Retrieval  
29 March–2 April 2026, Delft, The Netherlands  
[\[paper, slides, website\]](#)
  - **Conversational Search: From Fundamentals to Frontiers in the LLM Era**  
Fengran Mo, **Chuan Meng**, Mohammad Aliannejadi, and Jian-Yun Nie  
**SIGIR 2025**: The 48th International ACM SIGIR Conference on Research and Development in Information Retrieval  
13–17 July 2025, Padua, Italy  
[\[paper, slides, website\]](#)
  - **Query Performance Prediction: Theory, Techniques and Applications**  
Negar Arabzadeh, **Chuan Meng**, Mohammad Aliannejadi, and Ebrahim Bagheri  
**WSDM 2025**: The 18th ACM International Conference on Web Search and Data Mining  
10–14 March 2025, Hanover, Germany  
[\[paper, slides\]](#)

- **Query Performance Prediction: Techniques and Applications in Modern Information Retrieval**  
Negar Arabzadeh, **Chuan Meng**, Mohammad Aliannejadi, and Ebrahim Bagheri  
**SIGIR-AP 2024**: The 2nd ACM SIGIR-AP conference  
9–12 December 2024, Tokyo, Japan  
[\[paper\]](#), [slides](#)
- **Query Performance Prediction: From Fundamentals to Advanced Techniques**  
Negar Arabzadeh, **Chuan Meng**, Mohammad Aliannejadi, and Ebrahim Bagheri  
**ECIR 2024**: The 46th European Conference on Information Retrieval  
24–28 March 2024, Glasgow, UK  
[\[paper\]](#), [slides](#)
- Lecturing: so far, I have delivered 3 lectures (1 course lecture and 2 guest lectures) in the United Kingdom, the United States, and the Netherlands.
  - **Conversational Search: From Fundamentals to Frontiers in the Agent Era** (guest lecture, about **50** MSc and final-year undergraduate students), The University of Edinburgh, March 2026  
Host: [Alexandra Birch](#)  
[slides](#)
  - **Conversational Search: From Fundamentals to Frontiers in the LLM Era** (guest lecture, about **40** undergraduate students), University of Massachusetts Amherst, October 2025  
Host: [Razieh Negin Rahimi](#)  
[slides](#)
  - **Recommender Systems** (course lecture, about **50** students), University of Amsterdam, June 2025  
Course design and delivery of the lecture “Sequential and Large Language Model-based Recommender Systems”  
[slides](#)
- Teaching assistance:
  - **Information Retrieval** (about **60** students), University of Amsterdam, 2022–2023; project design and grading
  - **Information Retrieval** (about **60** students), Shandong University, 2020; assignment and project design, and grading
  - **Natural Language Processing** (about **70** students), Shandong University, 2019; assignment and project design, and grading
- PhD mentoring:
  - [Paul Owoicho](#), The University of Edinburgh, with Dr. [Jeff Dalton](#)
  - [Lili Lu](#), Università della Svizzera italiana (USI), with Prof. dr. [Fabio Crestani](#)

## INVITED TALKS

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So far, I have delivered 10 invited talks in academia (the Royal Society, UCL, King’s College London, the University of Glasgow, the University of Sheffield, and the China Conference on Information Retrieval) and industry (Amazon in Seattle and London, and Objective, Inc.).

- **The Role of Information Retrieval in Deep Research**  
16 August 2026, Young Scholars Forum at the 32nd China Conference on Information Retrieval (CCIR 2026)  
Host: Information Retrieval Technical Committee, Chinese Information Processing Society of China  
[slides](#)
- **The Role of Information Retrieval in Deep Research**  
24 June 2026, King’s College London (KCL)  
Host: [Yulan He](#), professor at King’s College London
- **Revisiting Text Ranking in Deep Research**  
5 March 2026, University of Sheffield  
Host: [Nikolaos Aletras](#), professor at the University of Sheffield

- **Revisiting Text Ranking in Deep Research**  
3 March 2026, the Royal Society
- **Opportunities and Challenges of LLMs in Information Retrieval**  
21 October 2024, University of Glasgow  
Host: Iadh Ounis, professor at the University of Glasgow  
[\[slides\]](#), [\[post\]](#)
- **Predicting the Right Moment for System Initiative in Mixed-Initiative Conversational Search**  
26 August 2024, Amazon (London)  
Host: Gabriella Kazai, principal applied scientist at Amazon  
[\[slides\]](#)
- **Opportunities and Challenges of LLMs in Information Retrieval**  
14 August 2024, Objective, Inc.  
Host: Pablo Mendes, co-founder & CEO at Objective, Inc.  
[\[slides\]](#)
- **Opportunities and Challenges of LLMs in Information Retrieval**  
17 April 2024, Amazon (Seattle)  
Host: Shervin Malmasi, applied science manager at Amazon  
[\[slides\]](#)
- **System Initiative Prediction and Query Performance Prediction for Conversational Information Seeking**  
3 November 2023, University College London (UCL)  
Host: Xi Wang, lecturer at the University of Sheffield  
[\[slides\]](#), [\[post\]](#)
- **Query Performance Prediction for Conversational Search**  
18 May 2023, University of Glasgow  
Host: Iadh Ounis, professor at the University of Glasgow  
[\[slides\]](#), [\[post\]](#)

## ADMINISTRATIVE ACTIVITIES

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- Manager for the [LinkedIn account](#) of IRLab (Information Retrieval Lab, University of Amsterdam), 2025
- Webmaster for the [IRLab website](#), University of Amsterdam, 2023–2024
- Chair for internal seminars at IRLab, University of Amsterdam, 2023
- Lead organizer of the IRLab BBQ event, University of Amsterdam, 2023

## RESEARCH FUNDING PROPOSALS

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I have contributed to the methodology design of 3 national research funding proposals in China.

- **National Key Research and Development Program of China** (Project No.: 2020YFB1406700)  
Contributor (methodology design)  
Development of an AI-assisted writing robot system based on news reporting scenarios
- **National Natural Science Foundation of China (NSFC)** (Project No.: 61972234)  
Contributor (methodology design)  
Research on Explainable Medical Health Question-Answering Technology Based on Multi-Source Heterogeneous Data Fusion
- **National Natural Science Foundation of China (NSFC)** (Project No.: 62072279)  
Contributor (methodology design)  
Research on Conversational Recommendation Systems and Functional Enhancements Based Primarily on Implicit Feedback Data



## SCHOLARSHIPS & AWARDS

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- Excellent Master's Thesis of Shandong Province, 2022
- National Scholarship (China), 2020/2016
- Outstanding Graduates of Shandong Province, 2021/2017
- SIGIR Student Travel Grant, 2020
- Scholarship for Outstanding Postgraduate Cadres of Shandong University, 2020/2019
- AAAI Student Scholar Scholarship, 2019
- Academic Scholarship for Master Students of Shandong University, 2019
- Outstanding Students of Shandong Province, 2017

## RESOURCES

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My open-source code and data have attracted over **800** GitHub stars and over **14,000** Hugging Face downloads, and my code has been reused by companies including Snowflake and Dataminr. I have curated the following resources:

- a [resource repository](#) on Hugging Face (**over 14,000 downloads**) providing the passage corpus, pre-built retriever indexes, and agent execution trajectories for deep research.
- a [code repository](#) (**over 200 stars**) providing scripts for text ranking in deep research, within a Python/PyTorch framework.
- a [code repository](#) (**over 100 stars**) providing scripts for fine-tuning open-source LLMs to generate relevance judgments, within a Python/PyTorch framework.
- a [code repository](#) (**over 110 stars**) providing a comprehensive implementation of query performance prediction (QPP) methods, within a unified Python/PyTorch framework.
- a [code repository](#) (**over 50 stars**) offering a comprehensive implementation of ranked list truncation methods, within a unified Python/PyTorch framework.
- a [paper reading list](#) (**over 250 stars**) on knowledge-grounded dialogue systems.

Also, I have contributed to key open-source toolkits in information retrieval, including:

- [Pyserini \(v0.19.2\)](#), a Python toolkit for reproducible information retrieval research.
- [Tevatron](#), a flexible and efficient Python toolkit for training and inference of neural retrieval models.

## REFERENCES

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Dr. [Jeff Dalton](#)

Turing AI Acceleration Fellow

Chancellor's Fellow (Associate Professor) at The University of Edinburgh

[jeff.dalton@ed.ac.uk](mailto:jeff.dalton@ed.ac.uk)

Prof. dr. [Maarten de Rijke](#)

ACM Fellow

Distinguished University Professor at the University of Amsterdam

[m.derijke@uva.nl](mailto:m.derijke@uva.nl)

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Assistant Professor at the University of Amsterdam

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Dr. [Zhaochun Ren](#)

Associate Professor at Leiden University

[z.ren@liacs.leidenuniv.nl](mailto:z.ren@liacs.leidenuniv.nl)

Prof. dr. [Fabio Crestani](#)

Full Professor at Università della Svizzera italiana (USI)

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